

Experience of simple casting using Tin

Daisuke Kuroda

Dept. of Materials Science and Engineering

What is casting and castings ?

Casting is a one of manufacturing technique for industrial products made from metallic materials.

Molten metal is poured into a mold and allowed to solidify to create a product of the desired shape.

The word "casting" in English is pronounced "Cyuzou" in Japanese language.

Products produced by casting are called "Castings".

The great buddha at Todaiji temple in nara prefecture is the largest castings in the world.





Molten metal

Sand mold

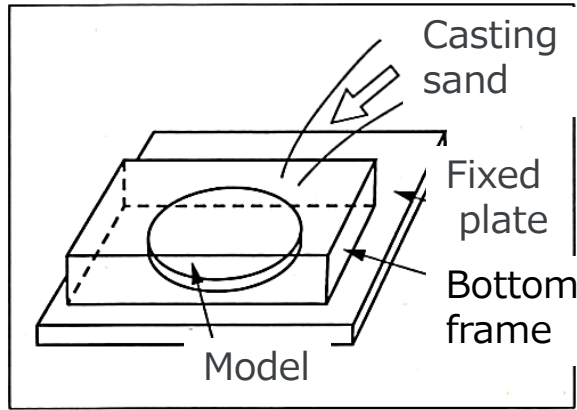
Molten metal is poured into a mold and allowed to solidify to create a product of the desired shape.



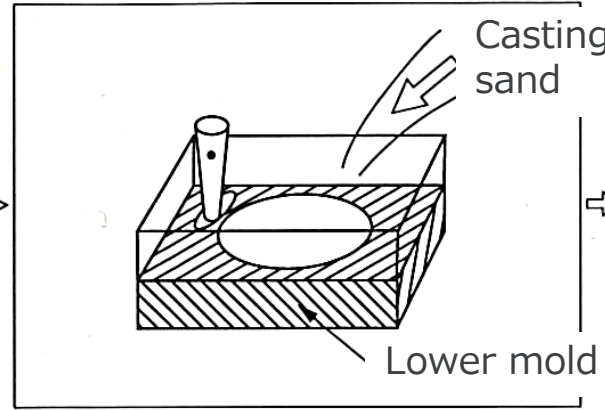
Automobile engines are a prime example of castings in modern industry. Creating the fins on the surface for heat dissipation by cutting or carving them out of a block of metallic material requires an enormous amount of effort and cost, but with casting, all you have to do is pour the material into a mold. What's more, the internal cavities that curve and branch out freely can only be created by casting.

Sand casting procedure

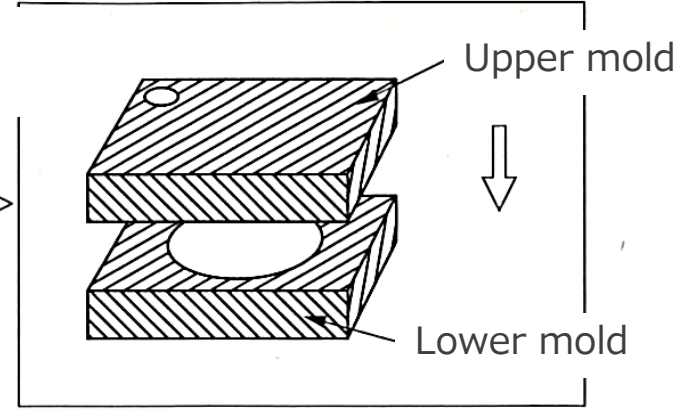
Model making $\Rightarrow$ Mold making $\Rightarrow$ Casting $\Rightarrow$ Removal of the castings



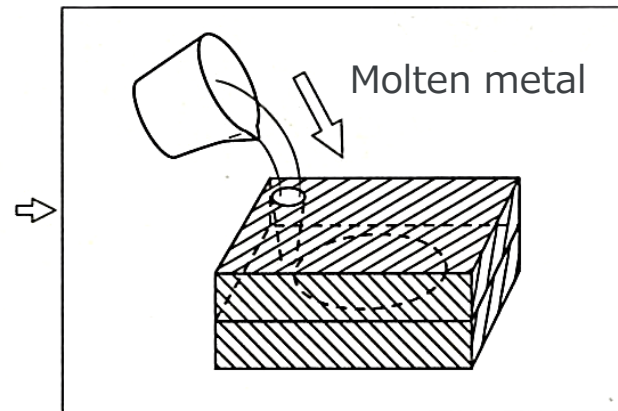
Making the lower mold



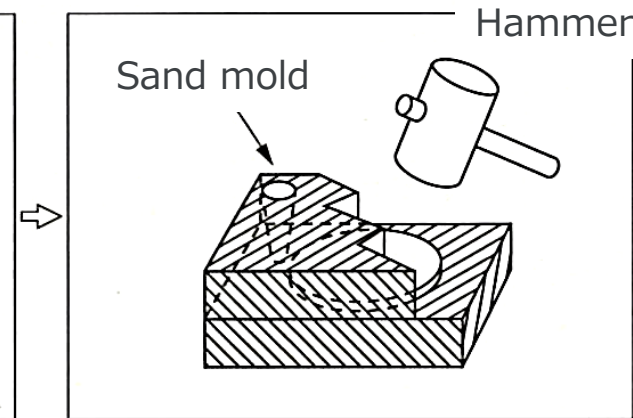
Making the upper mold



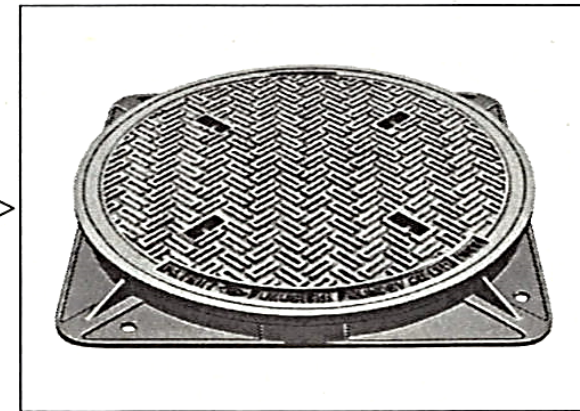
Align the upper and lower mold



Pouring molten metal



Dismantling the sand mold



Completion of the castings
(manhole cover)

Advantages and disadvantages of casting

Advantages of casting

1. Since it is shaped with liquid, complex shapes can be easily made.
2. The same shape can be mass-produced at the same time.



Casting cavity

Disadvantages of casting

1. It shrinks during the cooling process, so it is difficult to get accurate dimensions.
2. Not good at thin parts, extremely fine parts, or parts with sharp edges.
3. Pores called "Casting cavity" tend to form inside.
4. There are limitations to the surface finish.
5. Special equipment is required.



Cast surface

What is pewter ?

Pewter is an alloy made by adding antimony and copper to tin. Its composition varies, but standard pewter contains 91% tin (Sn), 7% antimony (Sb), and 2% copper (Cu). In the past, pewter was sometimes made with lead (Pb), but now it is common to use pewter without lead, because lead is harmful.

Pewter castings have a beautiful silver-white metallic luster, and are flexible and strong enough to be used to make ornaments such as vases and statues, tableware, and accessories.

Pewter can also be easily melted using a household stove and enamel pan, making it an ideal material for enjoying casting at home.



Enjoy casting with Tin !

In this time, we will use a method called "Fukinagashi," in which a depression in the shape you want is made in the casting sand and molten metal is poured onto it.

This is the most basic casting method, but the casting sand we use is the real thing, the same sand we use at our college.

We hope you enjoy the texture of the casting.



Place the ring on the board and place the model you want to make inside it.



Add casting sand and pack it down firmly with your fingers.



Add more casting sand until it fills the ring and tamp it down firmly.



Cover with another board and turn the ring upside down.



Wet the casting sand around the model with water in order to easier to remove the model from the mold.



Use tweezers to grasp the handle attached to the model and remove it from the mold.



Once you have removed the model, give me the mold and I will pour the tin into it.



Make sure it has cooled completely before removing the casting from the mold. Rinse the casting sand with water.

Casting completed !

If you want to try casting yourself, try making a mold again.

✂ Be careful not to get burned !

Reports

- Additive manufacturing is becoming more commonplace as a new process for manufacturing metal products, replacing casting. What are some of the advantages and disadvantages of additive manufacturing?
- Please tell us your impressions of the simple casting experience using tin.